■ Alphawise: Comprehensive Feature-by-Feature Demo Script

Exhaustive Screenplay Covering Every Specialty, Tab, Formula, and Engine Mechanism

Target Video Duration: 6:30 – 7:30 Minutes | **Live App:** http://localhost:3000 | **Presentation Target:** Technical Judges & Evaluators

■ COMPLETE FEATURE INVENTORY COVERED IN THIS SCRIPT:
1. Top Header: Branding, Dark/Light theme toggle, BlackRock Tear Sheet PDF export, Live backend heartbeat.
2. Left Sidebar Universe Builder: 4 Presets (Tech, 60/40, All-Weather, Inflation), Equal Weight, Clear All, Custom Ticker Add (+SPY, +QQQ, +NVDA, +GLD, +TLT, +BTC), dynamic weight inputs, 100% budget constraint validation meter, Lookback window (1Y–5Y), US 10Y Rf input, Max Allocation Cap bounds (25%–100%).
3. Numerical Engine & Solver: Charnes-Cooper hyperbolic transformation, SciPy SLSQP & CVXPY quadratic solvers, Ledoit-Wolf shrinkage, 4 Top Scorecards (Sharpe, Volatility, Return, Shannon Eigenvalue Effective Bets Neff).
4. AI Diagnostic Hub: 100-point Multi-Factor Health Score & 4 component breakdown bars, Dual-Voice Commentary (Institutional CIO Fiduciary Memo vs. ■■ FinTwit Roast), Groq Llama 3.3 70B AI Re-roll.
5. 6 Tabbed Analytics Panes: Efficient Frontier curve with custom markers, Macro Crisis Stress-Test (5 historical crashes with downside cushion & recovery months), What-If Sandbox (GLD, TLT, BTC, Cash injection with live Sharpe delta), Pearson Correlation Heatmap, Individual Asset Volatilities bar chart, 63-Day Rolling Stress Volatility time-series.
6. Execution & Copilot: Rebalance Trade Execution Ticket with trade deltas Δw, Copy to Clipboard, 1-click Apply Weights, Floating AI Copilot with real-time portfolio awareness & suggestion chips.

1. Header Navigation, Theme Engine & Live Backend Health (0:00 – 0:40)

SCENE / TIMING	WHERE TO POINT & EXACT ACTION	WHAT TO SAY (SPOKEN SCRIPT & JUDGES EXPLANATION)
0:00 - 0:20 Terminal Identity & Problem Hook	• Screen starts full browser at http://localhost:3000. • Point cursor to top logo: Alphawise TERMINAL v2.5. • Hover over the subtitle tag: *'Institutional Quantitative Engine & AI Portfolio Intelligence'*.	"Judges, modern wealth management tools suffer from a fatal flaw: they treat diversification as a nominal ticker count, while standard Markowitz optimization in practice acts as an 'error-maximizer' that produces unstable allocations. This is **Alphawise Terminal v2.5**—an institutional platform combining Modern Portfolio Theory, spectral risk decomposition, historical crisis stress testing, and Groq-powered AI diagnostics into a unified financial terminal."
0:20 - 0:40 Theme Switcher & Heartbeat Check	• Point to the top-right **Live Status Badge**: pulsing green dot next to Healthy: Alphawise Quantitative Engine. • Click ■ **Theme Toggle** (switches to crisp daylight mode, then back to dark terminal mode).	"Starting in the top header: Alphawise features an active health monitor polling our FastAPI backend at /api/v1/health, confirming real-time connectivity to our numerical engine. Next to it is our persistent Theme Engine: one click toggles between institutional dark mode and high-contrast day mode, with zero layout flash or style lag."

2. Portfolio Universe Builder, Presets & Constraints (0:40 – 1:45)

SCENE / TIMING	WHERE TO POINT & EXACT ACTION	WHAT TO SAY (SPOKEN SCRIPT & JUDGES EXPLANATION)
0:40 - 1:05 Module 1: Strategy Presets & Quick Utils	• Move cursor to **Portfolio Presets** in the sidebar. • Point out the 4 presets: ■ Tech Megacaps, ■■ Classic 60/40, ■■ All-Weather, ■ Inflation Shield. • Click ■ Tech Megacaps. • Point to ■■ Equal Weight and ■ Clear All utility buttons.	"In the left control panel, Module 1 provides one-click institutional templates: • **Tech Megacaps** (AAPL, MSFT, NVDA, GOOGL, AMZN) • **Classic 60/40** (Equities and Treasuries) • **All-Weather** (Risk parity across bonds, commodities, and equities) • **Inflation Shield** (TIPS, Energy, Gold). Let's select Tech Megacaps. Notice our quick utility buttons: **Equal Weight** rebalances any portfolio uniformly, while **Clear All** resets the canvas instantaneously."

1:05 - 1:25 **Module 2: Ticker Add & Allocation Meter**	• Point to Add Asset bar. • Click Quick Add chips: +GLD (Gold) and +TLT (Treasuries). • Point to holdings table rows (editable weight % inputs & delete buttons). • Point to **Allocation Meter**: green track reading **100.0% Balanced**.	"In Module 2, users can add any stock, ETF, or crypto ticker. We also provide Quick-Add chips for standard hedging assets: let's click **+GLD** and **+TLT**. Each asset features inline editable target weights. Crucially, look at the **Allocation Meter**: it enforces an exact budget constraint (sum w_i = 100%). If weights diverge from 100%, the bar turns amber or red and disables execution, guaranteeing mathematical consistency."
1:25 - 1:45 **Module 3: Quantitative Parameters**	• Scroll down sidebar to Module 3. • Point to **Historical Lookback Window** (1Y, 2Y, 3Y, 5Y). • Point to **Risk-Free Rate (Rf)** (set to 4.5%). • Point to **Max Allocation Cap** (No Cap, 50%, 40%, 30%, 25%).	"Module 3 controls the mathematical bounds of our optimizer: 1. **Lookback Window:** Selects the historical return sampling period (1 to 5 years). 2. **Risk-Free Rate:** Set to the US 10-Year Treasury benchmark (4.5%) to anchor the Sharpe denominator. 3. **Max Allocation Cap:** This introduces inequality box constraints (w_i <= w_max), preventing the solver from creating extreme, un-investable corner portfolios in a single asset."

3. Optimization Execution & Top Performance Scorecards (1:45 – 2:40)

SCENE / TIMING	WHERE TO POINT & EXACT ACTION	WHAT TO SAY (SPOKEN SCRIPT & JUDGES EXPLANATION)
1:45 - 2:10 Optimizer Execution & Metric Cards 1 & 2	• Click ■ Run Institutional Optimization. • Highlight the ~800ms solver execution. • Point to **Card 1: Optimal Sharpe Ratio** (green delta badge). • Point to **Card 2: Annualized Volatility (σp)**.	"Now, let's execute our solver. Alphawise runs the Charnes-Cooper hyperbolic transformation to convert the non-linear Maximum Sharpe problem into a convex quadratic program solved via SciPy SLSQP and CVXPY. In Card 1, see the **Optimal Sharpe Ratio**: it highlights the exact delta improvement over the user's current baseline. Card 2 displays the portfolio's **Annualized Volatility (σp = sqrt(w^T Σ w))**, factoring in the full 252-day covariance structure."
2:10 - 2:40 Card 3 & Card 4: Spectral Bets (N_eff)	• Point to **Card 3: Expected Return (μp)**. • Point to **Card 4: Effective Bets (N_eff)** and hover over INFO ■.	"Card 3 displays the annualized compounded return. Now examine our fourth card: **Effective Number of Bets (N_eff)**. This is a signature quantitative feature. Standard brokers claim that owning 7 stocks gives you 7 bets. Alphawise performs a full spectral decomposition of the Pearson correlation matrix (C = V Λ V^T) and computes the Shannon entropy of normalized eigenvalues: N_eff = exp(-Σ p_k ln p_k). This mathematically proves that our 7 tech and growth holdings only yield ~2.7 truly independent risk factors. We replace nominal diversification with orthogonal factor exposure."

4. AI CIO Diagnostic Hub: Health Score & Dual Commentary (2:40 – 3:45)

SCENE / TIMING	WHERE TO POINT & EXACT ACTION	WHAT TO SAY (SPOKEN SCRIPT & JUDGES EXPLANATION)
2:40 - 3:15 Health Score & 4 Factor Breakdown Bars	• Scroll to **Section 2: AI Diagnostic Hub**. • Point to the large **Health Score (e.g. 84/100)** and grade badge. • Trace the 4 sub-metric progress tracks: 1. Diversification 2. Sharpe Efficiency 3. Volatility Drag 4. Tail Resilience.	"Section 2 is our **AI Portfolio Diagnostic Hub**. At the top is our composite 100-point Health Score, synthesized across 4 quantitative pillars: 1. **Diversification:** Evaluates our spectral N_eff against nominal asset count. 2. **Sharpe Efficiency:** Current Sharpe relative to the theoretical tangency frontier. 3. **Volatility Drag:** Penalizes excessive variance compounding drag. 4. **Tail Resilience:** Evaluates 95% Cornish-Fisher Value-at-Risk accounting for skewness and kurtosis."
3:15 - 3:45 Institutional Memo vs. ■■ Roast & AI Re-roll	• Point to **Executive Fiduciary Summary** and Callout Cards in Institutional Memo. • Click ■■ Roast My Portfolio button. • Highlight the flaming pill: ■■ Extreme Roast & read the punchline. • Point to ■ AI Re-roll button.	"Alphawise bridges complex math to human understanding through dual-persona AI powered by Groq Llama 3.3 70B. In **Institutional Memo** mode, the AI writes a fiduciary investment teardown analyzing factor orthogonality, tail risks, and rebalance theses. Click **Roast My Portfolio**, and the persona shifts to a sharp, satirical FinTwit breakdown—humorously tearing down retail biases and hype-chasing in viral language. Users can click **AI Re-roll** to query the Groq LLM endpoint dynamically for fresh perspectives."

5. Interactive Visual Analytics & Crisis Stress-Testing (3:45 – 5:15)

SCENE / TIMING	WHERE TO POINT & EXACT ACTION	WHAT TO SAY (SPOKEN SCRIPT & JUDGES EXPLANATION)
3:45 - 4:10 **Tab 1: Efficient Frontier Visualization**	• Scroll to **Section 3: Tabbed Visualizations**. • Click Tab 1: ■ Efficient Frontier. • Hover cursor over: - Frontier curve (scatter line) - Green star: Tangency Max Sharpe - Blue triangle: Min Volatility - Red diamond: Current Portfolio - Gray circles: Individual assets.	"Section 3 provides 6 deep quantitative visualization panes. Pane 1 renders the **Efficient Frontier**. We compute and plot the parametric Markowitz hyperbola. Notice our visual anchors: • **Green Star:** The Tangency Portfolio maximizing the Sharpe angle from Rf. • **Blue Triangle:** Global Minimum Variance portfolio. • **Red Diamond:** The user's current baseline. • **Gray Points:** Individual asset coordinates showing their standalone risk-return profiles."
4:10 - 4:35 **Tab 2: Macro Crisis Stress-Test Simulator**	• Click Tab 2: ■ Macro Crisis Stress-Test. • Click scenario selector pills: 1. **2020 COVID Shock** 2. **2022 Rate Hike** 3. **2008 GFC** 4. **1970s Stagflation**. • Point to **Downside Cushion Badge (+8.4%)**, Drawdown comparison boxes, and worst/best asset highlights.	"Pane 2 is our **Historical Crisis Stress-Test Simulator**. Static variance assumes normal Gaussian returns. We stress-test the user's exact asset weights against 5 historical macro shocks: 2020 COVID, 2022 Fed rate tightening, 2008 Lehman collapse, and 1970s stagflation. Alphawise calculates drawdown comparisons in real time: showing our optimized portfolio provides an **8.4% downside capital cushion**, halves estimated recovery time, and highlights the worst-hit asset versus the best safe haven."
4:35 - 4:55 **Tab 3: Interactive What-If Sandbox**	• Click Tab 3: ■■ What-If Sandbox. • Click test injection pills: + 10% Gold (GLD), + 15% Long Treasuries (TLT). • Point to projected metrics: Sharpe Delta (+0.14), Volatility Delta (-1.8%), Bets Improvement (+0.7). • Point to ■ Inject into Portfolio Table button.	"Pane 3 is our **Interactive What-If Sandbox**. Before reallocating capital, risk managers can simulate injecting orthogonal asset classes like physical gold, long-term bonds, or Bitcoin. It computes the marginal Sharpe delta, volatility reduction, and effective bets improvement in real time, and allows users to click **Inject into Portfolio Table** to apply it with zero friction."
4:55 - 5:15 **Tabs 4, 5 & 6: Heatmap, Bar Chart & Rolling Vol**	• Click Tab 4: ■■ Correlation Matrix (point to colors). • Click Tab 5: ■ Asset Volatilities (point to bar comparisons). • Click Tab 6: ■ Rolling Stress Volatility (point to 63-day time-series).	"Pane 4 displays the interactive Pearson Correlation Heatmap: identity diagonal in acid green, high collinearity in coral, and negative correlation shelters in emerald. Pane 5 compares individual asset volatilities against portfolio-level risk. Pane 6 plots **63-Day Rolling Realized Volatility** over historical trading days, proving that our optimized tangency portfolio consistently dampens volatility spikes during market turbulence."

6. Trade Execution Ticket, AI Copilot & Tear Sheet Export (5:15 – 6:30)

SCENE / TIMING	WHERE TO POINT & EXACT ACTION	WHAT TO SAY (SPOKEN SCRIPT & JUDGES EXPLANATION)
5:15 - 5:40 Rebalancing Ticket & 1-Click Execution	• Scroll to **Section 4: Rebalancing Table**. • Trace table headers: Asset, Current %, Optimized %, Delta (Δw), Action (BUY/SELL/HOLD). • Point to ■ Copy Trade Ticket. • Point to ■ Apply Optimized Weights.	"In Section 4, Alphawise converts quantitative optimization into concrete trading orders. Our **Rebalance Ticket** calculates the exact weight delta (Δw) for each security with explicit BUY, SELL, or TRIM actions. Users can click **Copy Trade Ticket** to copy broker-ready limit/market instructions to their clipboard, or click **Apply Optimized Weights** to overwrite the active universe parameters in one click."
5:40 - 6:05 Floating AI Copilot & State Awareness	• Click the floating button in bottom right: ■ Ask Copilot. • Show the slide-out drawer. • Point to the green active context label. • Click suggestion pill: *'Why buy/sell these?'*. • Show response streaming in.	"In the bottom right corner is our floating **AI Copilot**. The Copilot is state-aware: it continuously receives the full portfolio covariance matrix, current holdings, and trade deltas via structured prompt injection. Users can click pre-built prompt pills or type questions like *'Why are you recommending these trades?'* or *'Check valuation for NVDA'*, receiving instant, contextually accurate answers without leaving the dashboard."
6:05 - 6:30 BlackRock Tear Sheet & Final Pitch	• Scroll to the top header. • Click ■ Export Tear Sheet button. • Look directly into camera for concluding statement.	"Finally, by clicking **Export Tear Sheet** in the header, Alphawise triggers an institutional, BlackRock-grade print layout compiling executive metrics, holdings, and risk disclaimers into a client-ready document. To conclude: Alphawise delivers end-to-end portfolio intelligence—from spectral math and convex solvers to stress testing, AI commentary, and seamless trade execution. Thank you, judges. We are now ready for your questions!"

■ Technical Architecture & Math Reference for Judges' Q&A;

• Mathematical Formulation:
Max Sharpe Tangency Problem: min_w - (w^T μ - Rf) / sqrt(w^T Σ w) subject to Σ w_i = 1 and 0 <= w_i <= w_max.
Solved via Charnes-Cooper variable substitution y = w / (w^T μ - Rf) yielding a quadratic programming problem.

• Spectral Shannon Entropy:
Eigenvalues λ_1...λ_N of correlation matrix C. Normalized weights p_k = λ_k / N. Neff = exp(-Σ p_k ln p_k).

• Robustness to Overfitting:
Ledoit-Wolf shrinkage towards constant correlation matrix reduces estimation variance when T is small relative to N.

• Tech Stack & Performance:
FastAPI (Python 3.11), NumPy, Pandas, SciPy, CVXPY, Chart.js, Groq Llama 3.3 70B (sub-second generation).